



Imagin: Neeta Wadiker

in real time

a washing machine, which uses only specific cleaning routines.

Embedded systems are used in navigation tools like global positioning systems (GPS), automated teller machines (ATMs), networking equipment such as routers and switches, digital videos and cameras, mobile phones, aerospace applications, telecom applications, etc. Even toys make use of embedded systems (see box, 'What goes into an embedded system?').

Embedded systems are usually low in cost and are easily available off the shelf for most applications. They usually have low design risks, since it is easy to verify the designs using evaluation boards. The easy availability of good design tools (many of them in the freeware domain) and software engineers have been two key factors in fuelling the growth of embedded systems.

Embedded systems have received a major shot in the arm over the past two or three years due to three factors. The first was the development of standard run-time platforms like Java, which enabled their use in myriad ways that

were unimaginable in the past. The second factor was the emergence of several integrated software environments, which simplified the development of these applications. The third factor was the coming together of embedded systems and the Internet, which made possible the networking of several embedded systems to operate as part of a large system across networks—be it a LAN, WAN or the Internet. This convergence of embedded systems with the Internet is going to transform the way we live.

In the driving seat

There are several tasks in which real time OSes beat their desktop counterparts hands-down. A common application of embedded systems in the real world is in automobiles because these systems are cheap, efficient and problem-free. Almost every car that rolls off the production line these days makes use of embedded technology in one form or the other. RTOSs are preferred in this area due to their fast response times and minimal system requirements.